Heritage Bank & Trust — Enterprise Domain Context

About the Organization

Heritage Bank & Trust is a mid-sized regional bank with **$48 billion in assets**, operating **210 branches** across the Midwest and Southeast US. Founded in 1923, Heritage serves approximately **1.2 million retail customers** and **85,000 business banking clients**. Core banking operations run on **IBM AS/400 (IBM i / FiServ Precision)** — a platform that has been in continuous production for 31 years and is treated as the authoritative source of record for all accounts, transactions, and interest calculations.

Heritage Digital is the internal modernization programme that builds new customer-facing and staff-facing digital channels while leaving the proven AS/400 core intact. The programme follows a **strangler-fig pattern**: a **Banking Integration Hub (BIH)** adapts between modern REST microservices and the AS/400, progressively replacing screen-scraping and batch integrations one domain at a time.

Two product teams operate under Heritage Digital:

- **Retail Banking Team** — consumer mobile and online banking for 1.2 million retail customers
- **Commercial Banking Team** — SMB business portal and commercial payments for 85,000 business clients

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Enterprise User Personas

Persona	Team	Description
Retail Customer	Retail	Individual account holder; manages personal accounts digitally
SMB Owner	Commercial	Small-medium business banking client; multi-account access, ACH payroll, wires
SMB Authorized User	Commercial	Employee granted access by SMB Owner; role-scoped (view/initiate/approve)
Branch Teller	Both	In-branch staff; handles transactions on behalf of customers
Branch Manager	Both	Supervises tellers; handles exceptions; account opening authority
Customer Service Agent (CSA)	Both	Phone/chat support; read-only customer view; raises service tickets
Fraud Analyst	Both	Monitors transaction patterns; manages alert queue; card block/unblock
Compliance Officer	Both	BSA/AML reporting; SAR filing; audit trail access

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Core Banking System — IBM AS/400

- **Core system**: FiServ Precision running on IBM i 7.4 (Power9 hardware)
- All account records, transaction ledger, interest calculations, and GL entries live in AS/400

- **Batch processing windows** (critical constraint — no writes to AS/400 during these periods):
  - **Nightly batch**: midnight–4 AM CT — ACH settlement, interest posting, statement generation
  - **End-of-day (EOD)**: 6 PM CT — balance cutoff for same-day transactions
  - **Intra-day ACH returns**: processed at 10 AM and 2 PM CT
- **Real-time balance**: AS/400 provides available balance (after holds) via FiServ Open Solutions API; ledger balance updates at EOD batch
- **AS/400 is authoritative**: no digital service may show a balance or transaction status that contradicts AS/400 without flagging it as provisional

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Enterprise Account Types

Account	Description
Checking (DDA)	Demand deposit account; standard and interest-bearing variants
Savings	Regulation D compliant; 6 withdrawal limit per statement cycle (legacy rule enforced)
Money Market	Tiered interest; cheque-writing capability
CD (Certificate of Deposit)	Fixed term (3, 6, 12, 24, 36 months); early withdrawal penalty
HELOC	Home Equity Line of Credit; draw/repayment cycle; collateral in AS/400
Personal Loan	Installment loan; amortization schedule from AS/400 Loan module
Business Checking / Savings	Multi-signatory; ACH origination enabled

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Enterprise Compliance & Regulatory Framework

BSA/AML

- **Suspicious Activity Reports (SARs)**: filed via FinCEN BSA E-Filing; triggered by NICE Actimize case + Compliance Officer approval
- **Currency Transaction Reports (CTRs)**: auto-generated for cash transactions ≥ $10,000 in a single day
- Transaction monitoring: NICE Actimize (on-prem); receives real-time transaction feed via Kafka

SOX (Sarbanes-Oxley) — IT General Controls

- All production changes require **CAB (Change Advisory Board) approval** and a rollback plan
- **No Friday deployments** after 15:00 CT
- All customer data access events logged to the Audit Service (immutable, 7-year retention)

Regulation E

- Customers have the right to dispute unauthorized electronic fund transfers
- Heritage must provide provisional credit within 5 business days of a debit card dispute
- Resolution within 10 business days; written resolution notice required

PCI DSS SAQ-D

- Heritage handles cardholder data for debit card programmes (via FIS Worldpay)
- Annual PCI audit; quarterly network scans; pen test annually

Regulation D

- Savings accounts: maximum 6 withdrawals per statement cycle enforced at AS/400; Heritage must display this limit digitally and prevent/warn on excess

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Enterprise Integration Landscape

System	Purpose	Primary Team
IBM AS/400 (FiServ Precision)	Core banking — authoritative source of record	Shared / BIH
Banking Integration Hub (BIH)	Adapter layer between microservices and AS/400	Shared platform
FIS Worldpay	Debit card processing and authorization	Retail
EWS / Zelle	Peer-to-peer real-time payments	Retail
TCH RTP	Real-Time Payments network	Retail + Commercial
Fiserv CheckFree	Bill pay disbursement	Retail
NICE Actimize	Transaction fraud monitoring and AML	Both
Blend	Consumer account and loan origination platform	Retail
IBM MQ	Async message exchange between BIH and AS/400	Shared / BIH
Okta	Identity and access management (OIDC + SAML)	Both
Salesforce Financial Services Cloud	Service cases linked to customer accounts	Both
FedACH	ACH payment submission	Commercial
Fedwire / SWIFT	Domestic and international wire transfers	Commercial